

Remember Me: Virtual Reality for Alleviating Grief

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Figure 1: "Remember Me" scenes: (a) Story Island scene; (b) Meditation Island scene.

ABSTRACT

"Remember Me" is a Virtual Reality (VR) experience designed to assist individuals in coping with grief. It is grounded in the Dual Process Model and constructivism theory and inspired by interviews with grieving individuals and experts. The experience incorporates three main functions: listening to coping grief stories, flower planting, and grief meditation. A prototype evaluation with 48 participants demonstrated positive feedback regarding the experience design. Results indicated that participants found the experience meaningful and relaxing while suggesting improvements for clearer guidance.

Index Terms: Virtual Reality, Grief Support, Digital Memorial, Interactive Experience.

1 PROJECT URL

<https://www.rmvr.link/>

2 DESCRIPTION OF ARTWORK

"Remember Me" is a VR experience that aims to support individuals through their grief journey. Grounded in grief theory and informed by interviews with grieving individuals and experts, the experience provides virtual environments and interactive elements where users can engage with grief stories, participate in memorial activities, and find moments of peaceful reflection.

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2.1 Design Concept

The loss of a loved one is a profoundly traumatic event that will disrupt an individual's "assumptive world" by shattering a person's belief and sense of the meaning of the world [1]. The Dual Process Model is a fundamental framework for how individuals cope with bereavement. This approach describes that grief individuals face loss-oriented coping and restoration-oriented coping [3]. Loss-oriented coping means grief activities that connect an individual emotionally to the deceased, like crying and longing; on the other hand, restoration-oriented coping means accepting the life without the deceased and establishing new norms and identities [3]. The constructivist approach argues that reconstructing a world of meaning is central to the grief process[2]. Different from "letting go", maintaining a continued symbolic bond with the deceased is to reconstruct the identity of the relationship, which has a potential benefit for individuals [2]. It includes how people can reconstruct their grief narratives and how to help them relax and symbolically reconstruct the relationship to maintain a spiritual, continuing bond.

2.2 User and Expert Interviews

To identify the core functionalities of "Remember Me," this study conducted one-hour semi-structured interviews with two licensed therapists and three bereaved individuals. The interviews yielded five key insights:

- Participants find redirecting their focus towards grief helpful but suggest that directly facing death and life's nature could be more beneficial.
- Maintaining continuing bonds through items like photos and personal remnants is crucial for survivors.
- Grief meditation can be beneficial throughout different stages of the grieving process.
- Creating connections between individuals who have experienced similar losses and encouraging story sharing can be helpful.
- It is advisable to avoid offering this experience to those who have recently experienced grief.

Based on these insights, three main functions were developed for the VR experience: listening to coping grief stories, flower planting, and grief meditation.

2.3 User Journey

The whole experience of “Remember Me” contains two island scenes: Story Island and Meditation Island, designed with natural elements to foster a comfortable atmosphere throughout the user journey. Grounded in the Dual Process Model, Story Island primarily focuses on grief work, helping individuals understand that loss is a natural and common experience. Meanwhile, Meditation Island is designed to assist users in reconstructing meaning and facilitating relaxation.

Upon receiving a gentle reminder that the content may evoke sadness, the user enters Story Island. This scene is set at night, surrounded by a beautiful tree and grass. Floating candles around the user serve as memorial representations. When the user touches a candle, it lights up and plays one of the grief stories collected by “Remember Me” researchers. After fully interacting with the candle, the candle flies away to the sky as a memorial ritual.

Subsequently, the user is directed to Meditation Island. The user can select a flower seed to represent their loved one. Following this, the user is guided to a meditation area where they relax under the guidance of a professionally authorized grief meditation audio. Following the constructivist approach, the audio guides the user to relax, welcomes the beloved guest into their mind, and facilitates the reconstruction of the relationship with the deceased in the spiritual realm. Finally, after the meditation, the user can observe the flower they planted earlier in bloom.

2.4 Prototype Implementation

A 20-minute demo was developed for user testing, involving 48 participants who had experienced grief. Participants engaged with the demo using Oculus 2 headsets. They completed a questionnaire and participated in semi-structured interviews to provide feedback. The questionnaire employed a six-point likert scale to evaluate participants’ satisfaction with various aspects of the experience, including scene, audio, guide, and function, as well as their levels of relaxation. Before their participation, all individuals provided informed consent.

The results demonstrate that most participants were satisfied with the design of the experience. The scene satisfaction score (Mean = 4.94, SD = 1.019) and the audio satisfaction score (Mean = 5.17, SD = 0.9) were higher than the action guide satisfaction score (Mean = 4.27, SD = 1.067). Additionally, the results indicated good levels of relaxation among participants (Mean = 5.02, SD = 0.887) and found the experience to be meaningful in addressing grief (Mean = 4.85, SD = 1.167).

During the interview, the candle interaction received the most positive feedback, with some participants noting that it reawakened memories of their loved ones. They found resonance when listening to others’ stories. The symbolism of the candles flying away was appreciated as an effective metaphor for sending a wish to their deceased loved ones. Additionally, the meditation component of the experience also received positive feedback; most participants were able to follow the instructions and felt calm and at peace during the meditation. However, some participants suggested that the action guidance provided within the experience could be clearer. Few participants also expressed that they tried to avoid the grief-related stories when interacting with candles and recommended including more gentle reminders throughout the experience to better support users emotionally.

3 ARTIST INFORMATION

Yao Chen is a multimedia designer and researcher exploring the intersection of technology and humanity. She holds a Master of

Science in Multimedia and Entertainment Technology from The Hong Kong Polytechnic University (PolyU) and currently serves as a Research Associate at Research Centre for Cultural and Art Technology. Her research focuses on data visualization, WebXR, and AI-generated video design.

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REFERENCES

- [1] J. Gillies and R. A. Neimeyer. Loss, grief, and the search for significance: Toward a model of meaning reconstruction in bereavement. *Journal of Constructivist Psychology*, 19(1):31–65, 2006. doi: 10.1080/10720530500311182 1
- [2] R. A. NEIMEYER. Narrative strategies in grief therapy. *Journal of Constructivist Psychology*, 12(1):65–85, 1999. doi: 10.1080/107205399266226 1
- [3] M. Stroebe and H. Schut. The dual process model of coping with bereavement: A decade on. *OMEGA - Journal of Death and Dying*, 61(4):273–289, 2010. PMID: 21058610. doi: 10.2190/OM.61.4.b 1